Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Python backend systems**, **RESTful APIs**, and **cloud-native applications** across AWS and Azure environments. Experienced with CI/CD pipelines, PostgreSQL, microservices, and LLM-powered workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **Python backend services** for 500+ users using FastAPI and PostgreSQL, reducing API latency from **1.2s to 350ms** through query optimization, indexing strategies, and efficient database access patterns.
- Developed **ETL pipelines and data integrations** processing 5,000+ monthly jobs with validation checks, error handling, and CI/CD-backed releases, achieving 99%+ reliability in production environments.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **Flask APIs and React interfaces** handling 10,000+ daily requests, implementing RESTful endpoints for grading workflows while maintaining stable performance under peak instructor usage.
- Built **LLM-powered grading workflows** using GPT-4o and Claude APIs with OCR integration, reducing feedback turnaround by **60%** while tracking token usage and costs across AI model invocations.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **microservices and API integrations** using Python, AWS Glue, and S3, improving response times by **30%** while supporting distributed data workflows across production environments.
- Maintained **CI/CD pipelines and automated test suites** using Jenkins and PyTest, increasing build reliability by **68%** and enabling confident releases through unit and integration testing workflows.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed a **CRM platform with microservices architecture** using the MERN stack, implementing authentication, resource allocation, and billing services while syncing 2M+ records across locations.
- Automated **Docker-based CI/CD deployments** with Jenkins pipelines, cutting release time by **50%** and achieving 95%+ successful build validation across staging and production environments.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **data processing pipelines** across 100,000+ records using Python and PyTorch, enabling reproducible experimentation and high-confidence benchmarking for machine learning model evaluation.
- Produced **technical documentation and test scenarios** translating research findings into actionable guidance, improving cross-team alignment and engineering quality during prototype-to-production transitions.

Skills

Languages: **Python**, JavaScript, TypeScript, SQL, Java, C++, Bash
Backend & APIs: Flask, FastAPI, Django, Node.js, Express, RESTful APIs, microservices
Databases: PostgreSQL, MySQL, MongoDB, relational schema design, query optimization
Cloud & DevOps: AWS (S3, Glue, EC2), Azure, Docker, CI/CD, Jenkins, Git/GitHub
AI & Tools: OpenAI API, LLM integrations, Agile/Scrum, code reviews, pull requests

Projects

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Built a **containerized RESTful API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving deployment reliability and streamlining troubleshooting across production workflows.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, Docker*

- Built an **AI-powered SQL development workspace** supporting 10+ databases including PostgreSQL and MySQL, accelerating query authoring through schema-aware generation and unified database management.

PiXMod Live | *Next.js, MongoDB, Stripe, Transformers*

- Built an **AI-powered SaaS platform** for image editing using transformer models and MongoDB-backed workflows, implementing Stripe billing and improving premium conversion by **18%**.